



Additive Manufacturing of Glass

From Science to Applications

Additive Manufacturing Materials and Technologies

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Chapter Six - Laser-based approaches for 3D structuring of glass

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Abstract

This chapter gives a selected overview about laser-based approaches for generating three-dimensional (3D) structures in glass. A distinction is made between nanosecond and ultrashort pulsed lasers as well as CO₂ laser machining. While for nanosecond lasers linear absorption plays the dominant role for material ablation, ultrashort pulsed lasers are associated with nonlinear absorption processes offering both material ablation and modification; the latter may also be combined with chemical etching. In contrast, midinfrared CO₂ lasers exhibit high absorption, easily triggering thermal processes. Following these distinct absorption processes, selected processing strategies and applications for these different laser systems are discussed with a particular focus on hybrid approaches combining different laser systems or combining laser processing with chemical etching.



Keywords

Laser processing; Nanosecond; Femtosecond; Ultrashort pulsed; CO₂ laser; Modification; Ablation; Nonlinear processes

1. Introduction

Advancing the potentials of laser material processing by combining the laser approach with chemical etching has stimulated comprehensive research activities during the last decades, in turn fostering a variety of applications in, for example, optics, sensing, and lab-on-chip devices. While for these approaches often pulsed lasers are employed, a basic distinction can be made between the application of ultrashort laser pulses in a picosecond to femtosecond regime, lasers with nanosecond pulses, and cw-lasers with wavelengths in the midinfrared (M-IR) spectral region, such as CO₂ lasers.

From a very fundamental point of view, the specific absorption characteristics of dielectric materials, such as glass, have to be considered when applying such different laser sources. As being transparent in the visible spectral region (VIS), laser processing with nanosecond pulses can be achieved with lasers emitting in the ultraviolet (UV) region, such as excimer lasers. Different techniques such as laser-induced backside wet or dry etching have been demonstrated to decrease the ablation threshold of the material by adding a liquid or solid absorbent layer on the backside of the material.

While nanosecond laser processing is thermally driven due to heat deposited within the irradiated material, in turn causing melting and vaporization, the application of ultrashort laser pulses is often denoted as cold ablation, referring to the rather short pulse length as compared to the characteristic time constants for heat transfer to the lattice. As a result, machining quality using ultrashort pulses in terms of crack formation and melt generation is superior to the quality achieved by nanosecond lasers. Nonetheless, the quality attained with nanosecond lasers might be sufficient for certain applications with the advantage of significantly lower costs.

In contrast, CO₂ lasers with their M-IR emission wavelength of 10.6μm exhibit high absorbance in, for example, fused silica, causing a significant thermal heating of the irradiated material. Hence, thermal processes such as annealing or polishing are in favor of using this laser class.

Against this coarse yet fundamental background, in this chapter, selected laser processing approaches using different laser sources in conjunction with chemical etching are discussed with particular focus on glass processing.

2. Glass processing with ultrashort pulsed lasers

2.1. Nonlinear processes

Ultrashort pulsed lasers are classified by their pulse duration being in the femtosecond (10^{-15} s) and lower picosecond (10^{-12} s) regimen. Within the interaction time of the irradiated material with the laser, the photon energy of the laser is absorbed by the electrons of the material due to inverse Bremsstrahlung [1], [2], [3]. The low heat capacity of the electrons allows them to reach very high temperatures and thus generate a hot electron gas while the energy transfer from this gas to the lattice by electron-phonon interaction occurs on a timescale longer than the laser pulse duration [1], [2], [4], [5]. The hot electron gas transfers its energy to the lattice causing material removal, associated with a small heat-affected zone and thus high quality of the material removal [2], [4], [5]. Based on the different temperatures of the electron gas and the lattice, this ablation mechanism is typically described by a one-dimensional two-temperature model described by Eqs. (1), (2), (3), (4) [1], [6]

$$C_e \frac{\partial T_e}{\partial t} = -\frac{\partial Q(z)}{\partial z} - \gamma(T_e - T_i) + S \quad (1)$$

$$C_i \frac{\partial T_i}{\partial t} = \gamma(T_e - T_i) \quad (2)$$

$$Q(z) = -k_e \frac{\partial T_e}{\partial z} \quad (3)$$

$$S = I(t)A\alpha e^{-\alpha z} \quad (4)$$

where C_e and C_i are the heat capacities of the electron and the lattice subsystem, T_e and T_i are their temperatures, $Q(z)$ is the heat flux with z being perpendicular to the surface and k_e being the electron thermal conductivity, γ is the parameter characterizing

the electron-lattice coupling, and S is the laser heating source term with $I(t)$ being the laser intensity, A being the surface transmissivity, and α being the material absorption coefficient.

While for metals and small-band-gap materials where the energy of a single photon is sufficient to elicit linear absorption, transparent materials such as glasses do not permit this due to their large band gaps. Therefore, nonlinear processes are required to excite an electron from the valence band to the conduction band of the material, while a plasma critical density of about 10^{21} cm^{-3} for near-IR and VIS wavelengths is necessary to trigger ablation [3], [7]. At this density, the material becomes opaque due to the plasma frequency being the same as the laser frequency [3]. The electron excitation is described by a rate equation (Eq. 5), including multiphoton absorption, avalanche ionization, and relaxation of the generated electrons [7], [8].

$$\frac{\partial n(t)}{\partial t} = \alpha I(t)n(t) + \sigma_k I(t)^k - \frac{n(t)}{\tau} \quad (5)$$

where $n(t)$ denotes the excited electrons, α is the avalanche coefficient, $I(t)$ is the time-dependent intensity of the laser, σ_k is the k -photon absorption cross section for the smallest number of photons (k) to overcome the material band gap, and τ is the carrier decay time. The multiphoton absorption process is only dominant in weak electrical fields, while for strong electrical fields, tunnel ionization becomes the dominant process. Both regimes can be differed by the Keldysh parameter γ calculated by Eq. (6) [5], [9]

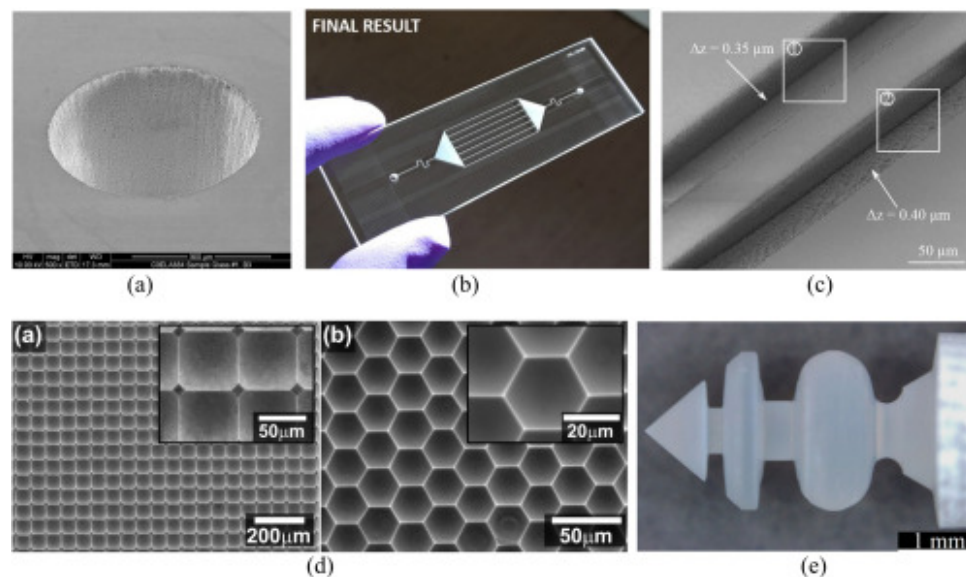
$$\gamma = \frac{\omega}{e} \sqrt{\frac{m_e c n \varepsilon_0 E_g}{I}} \quad (6)$$

where ω is the laser frequency, m_e is the effective mass of the electron, e is the fundamental electron charge, c is the speed of light, n is the linear refractive index, ε_0 is the permittivity of free space, E_g is the band gap of the material, and I is the laser intensity. For $\gamma \gg 1$, multiphoton absorption is the dominant process, while for $\gamma \ll 1$, tunnel ionization dominates [4], [5].

2.2. Direct ultrashort pulsed laser ablation

Material processing with ultrashort pulsed lasers is well known for its excellent machining quality due to the minimized heat-affected zone [1], [4], [5]. The ultrashort pulsed laser is feasible for direct ablation of all materials including large band-gap materials such as fused silica by direct irradiation. The single pulse ablation threshold for fused silica has been found to be 3.60 J/cm^2 for a 500fs laser having a wavelength of 1030nm [10]. Using more than a single pulse on one spot results in a decrease of the ablation threshold of the material due to incubation effects [11], [12]. The use of multipulse ablation by scanning the laser

across the surface to ablate cavities has shown an ablation efficiency of $0.41 \text{ mm}^3/(\text{Wmin})$ for fused silica applying a 230fs IR laser [13]. Splitting the individual pulses into GHz bursts with a separation of 460ps results in an increased ablation efficiency with increasing number of pulses per burst up to $3.05 \text{ mm}^3/(\text{Wmin})$ for a number of 10 pulses per burst [13]. Besides investigating the ablation behavior under ultrashort pulsed laser irradiation with different pulse sequences, common processes for glass processing such as drilling [14], [15] and cutting [15], [16] have been studied. Fig. 1A shows a hole drilled into 1 mm thick soda-lime glass using trepanning and a 750fs laser at a fluence of 26 J/cm^2 [14]. No microcracks or thermal damage is observed around the hole, confirming high-quality processing. Another application is the fabrication of microfluidic structures by direct laser ablation [17], [21]. Fig. 1B shows a microfluidic device fabricated with a 6ps laser in borosilicate glass [17]. Different ablative processes such as the generation of microfluidic channels, fabrication of through holes, and laser cutting (not shown in this device) are demonstrated. Besides these ablation-based processes, welding of two substrates is done with the same laser to close the microfluidic channels. This impressively demonstrates the versatility of the ultrashort pulsed laser. Fig. 1C shows microchannels fabricated with a 230fs laser on the rear side of fused silica with their sidewalls being rectangular due to the backside processing, which is not suitable for commonly used frontside ablation [18]. Besides the ablation of microfluidic structures, optical elements can be fabricated using the ultrashort pulsed laser for precise layerwise material removal and CO₂ laser polishing [22], microstructuring and CO₂ laser reshaping [23], or laser ablation of craters and subsequent etching [19]. The latter is shown in Fig. 1D for the fabrication of rectangular (left) and hexagonal (right) microlens arrays in fused silica where craters are directly ablated by a 30fs laser and subsequently processed by hydrofluoric (HF) etching [19]. Another process using ultrashort pulsed lasers is called laser turning for the fabrication of rotationally symmetric parts (Fig. 1D) out of fused silica using an 800fs laser [20].



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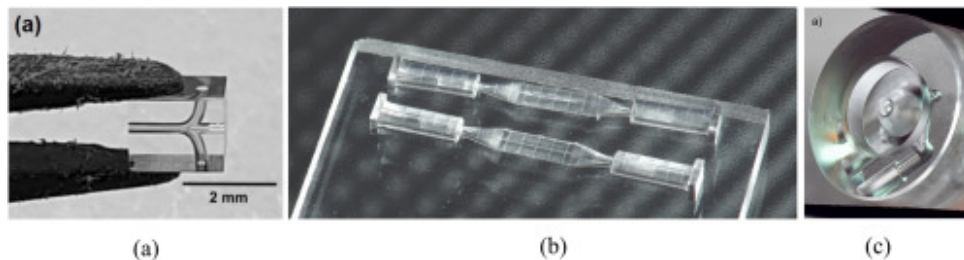
Fig. 1. Different processes used for direct ultrashort pulsed laser ablation of glasses (A) through hole drilled in soda-lime glass [14], (B) microfluidic device fabricated in borosilicate glass [17], (C) rectangular microchannels in fused silica fabricated by backside ablation [18], (D) microlens arrays in fused silica fabricated by laser ablation and wet etching [19], (E) laser turned fused silica [20].

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(E) Reprinted with permission J. Zettl, S. Rung, C. Esen, R. Hellmann, Tangential laser turning of fused silica using ultra-short laser pulses, J. Laser Micro/Nanoeng. 16 (1) (2021) 42–46. <https://doi.org/10.2961/jlmn.2021.01.2008> ↗, Japan Laser Processing Society.)

2.3. Selective laser-induced etching (SLE)

Fabricating 3D hollow structures inside transparent bulk materials is quite difficult. In 2001, Marcinkevičius et al. [24] demonstrated the fabrication of internal 3D structures inside silica glass by a two-step process consisting of a laser exposure and a wet etching step further referred to as selective laser-induced etching (SLE). Here, an ultrashort pulsed laser is applied to cause modification of the silica material having a higher etching rate compared to the unexposed material. The mechanism behind this process is described by a reduced bridging angle of the silica due to laser irradiation, resulting in an increased reactivity of the oxygen to the etchant [24]. By moving the laser spot, the desired 3D geometry is written inside the material and afterward removed by, typically, potassium hydroxide (KOH) [25], [26] or hydrofluoric acid (HF) [24], [27] etching. Using KOH instead of HF in an adequate solution has shown least saturation in elongated channels and thus a higher selectivity [28]. A similar behavior was found with KOH revealing a selectivity of 856 and 955 for channel and surface etching, being much higher as compared to 66 and 59 for HF, while the etching rates are 325 and 363 $\mu\text{m}/\text{h}$ with KOH compared to 320 and 287 $\mu\text{m}/\text{h}$ with HF, respectively [29]. Using KOH, a selectivity of about 1400:1 for fused silica with a corresponding etching rate of 290 $\mu\text{m}/\text{h}$ was reached [25]. Using a laser with two-pulse burst has shown an increased selectivity of 2000:1 for fused silica, while the etching rate is about 280 $\mu\text{m}/\text{h}$ for single-line horizontal channels and about 500 $\mu\text{m}/\text{h}$ for vertical planar structures [30]. Compared to the fast laser modification process, large structures can easily reach etching times up to several days [26], [31]. However, etching of several structured samples at once relativizes this time for mass production. Fig. 2 shows potential applications of this technology for the fabrication of (A) internal channels [32], (B) microfluidic devices [25], and (C) nested nozzles for biological cell breeding applications [31].



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Fig. 2. Different structures fabricated in glass by the SLE process (A) microfluidic device in [32], (B) microfluidic device [25], (C) nested nozzle in quartz glass [31].

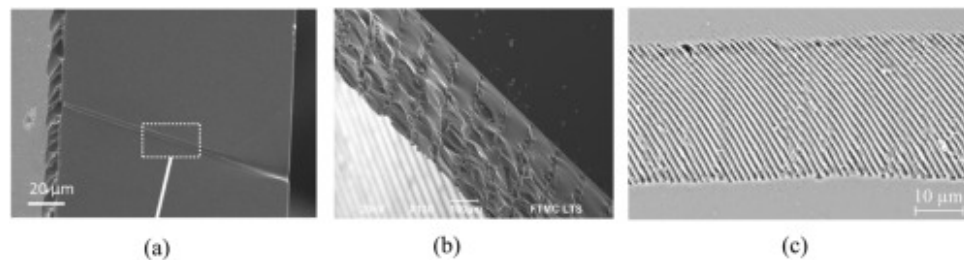
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2.4. Alternative ultrashort pulsed laser-based glass processing techniques

Besides direct laser ablation (Section 2.2) as well as selective-laser etching (Section 2.3), several other techniques and applications for transparent materials like glasses or optical polymers, such as writing as waveguide, welding, dicing, or microstructuring based on ultrashort pulsed lasers, have been demonstrated [33], [34], [35], [36], [37]. It has been shown that the irradiation of transparent material with ultrashort pulsed lasers can cause refractive index modifications [38]. Based on these results, functional waveguides have been specifically demonstrated in different glasses by an 800nm, 120fs laser [39]. The refractive index change in fused silica at the irradiated area is attributed to densified material showing an increased four- and three-membered ring structure in the glass substrate [40]. Even though ultrashort pulsed lasers are well known for their cold ablation, applying sufficiently high repetition rates can lead to the formation of waveguides due to heat accumulation [41]. This heat accumulation at high repetition rates can also be used for laser welding of transparent glasses with ultrashort pulsed lasers [36], [42]. However, low-repetition-rate lasers with 1 kHz have been used for welding of similar [43] and dissimilar transparent materials [44].

Another application using ultrashort pulsed lasers is the separation of glass by a so-called dicing process. Here, typically Bessel beams are formed out of Gaussian beams revealing an extended focal length as compared to Gaussian beams [45]. These beams can be used to generate voids in transparent materials by single laser shots revealing high-aspect ratios as shown for nanovoids with diameters of 200–400nm in fused silica and an aspect ratio of exceeding 1000:1 [46], respectively. Fig. 3 shows such a

microvoid generated in 150 μm thick borosilicate glass fabricated by a single 0.95 mJ and 50 fs laser pulse [47]. The authors also demonstrated voids in 1.5 mm thick substrates and mentioned no physical limit for samples of about 5 mm, exceeding theoretical aspect ratios of about 1200:1. Furthermore, modification tracks with an aspect ratio up to 3800:1 for a single pulse burst using a 19 ps laser were reported in fused silica [50]. By arranging several of such modifications side by side, a predetermined breaking point is introduced into the material, which is subsequently separated by induced stress to separate the glass substrates up to 1 cm thickness and more with beams revealing aspect ratios of >10.000:1 [33], [51]. Also, longer pulsed lasers with a pulse duration of 300 ps have been successfully studied [48]. Fig. 3B shows a 1 mm thick cleaved soda-lime substrate indicating the modifications generated by a 300 ps laser as well as the cracked area between them [48]. Furthermore, ultrashort pulsed lasers can be used to generate microstructures such as laser-induced periodic surface structures (LIPSS) on different glasses [52]. Two types of these structures are typically differentiated by their periodicity, namely, low-spatial-frequency LIPSS (LSFL) with their periodicity close to the laser wavelength and high-spatial-frequency LIPSS (HSFL) having a periodicity much smaller than the laser wavelength [53]. Fig. 3C shows LSFL on fused silica with their periodicity being somewhat smaller than the laser wavelength as being generated by a 230 fs laser using a Top-Hat beam shaper as well as an inclined angle of incidence [49], [54]. The most accepted theory describes the generation of these structures by the interference of the incident laser light with a surface scattered wave [55], [56].



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Fig. 3. (A) Microvoid generated in 150 μm thick borosilicate glass using a Bessel beam [47], (B) 1 mm thick soda-lime glass cleaved [48] and (C) LSFL on fused silica [49].

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<https://doi.org/10.1016/j.procir.2020.09.085> ↗, Elsevier, <https://creativecommons.org/licenses/by-nc-nd/4.0/> ↗. (C) Reproduced with permission S. Schwarz, S. Rung, C. Esen, R. Hellmann, Homogeneous low spatial frequency LIPSS on dielectric materials generated by beam-shaped femtosecond pulsed laser irradiation, *J. Laser Micro/Nanoeng.* 13 (2) (2018) 90–94. <https://doi.org/10.2961/jlmn.2018.02.0007> ↗, Japan Laser Processing Society.)

3. Glass processing with nanosecond pulsed lasers

3.1. Material ablation mechanism

Compared to material interaction with ultrashort laser pulses, where the temperatures of the lattice and the electrons differ strongly, causing nonthermal ablation, the ablation mechanism underlying longer laser pulses such as for nanosecond pulsed lasers differs. Here, both temperatures are the same, and thus, the ablation mechanism is thermally driven by heating, melting, and vaporization [1]. For these pulse durations, the material ablation threshold of dielectrics follows a $\tau^{1/2}$ behavior with τ being the pulse duration [7]. In this pulse length regime, thermal diffusion carries the energy out of the irradiated volume, while the thermal diffusion length can be calculated by $(\kappa\tau)^{1/2}$ with κ being the thermal diffusivity [5], [57]. Using short wavelengths and thus high-photon-energy lasers such as KrF, ArF, or F₂ excimer lasers with their wavelengths of 248, 193, and 157 nm, respectively, linear excitation of electrons to overcome the, for example, 9 eV band gap (138 nm) of fused silica is impossible [58]. While for short pulses multiphoton processes dominate the ablation of transparent materials, for longer pulses, that are, longer than a few tens of picoseconds, avalanche effects are predominant and the energy transfer to the lattice takes place on the range of the pulse duration [10], [57]. Thus, triggering the ablation process with long pulsed lasers, where the photon energy is less than the band gap of the material and no multiphoton absorption is triggered, initial seed electrons generated by, for example, thermal excitation or impurity and defect states are responsible for material ablation [57]. Furthermore, structural singularities or weak bulk absorption generated by the laser has been suggested [59]. However, also, two-photon processes have been reported for UV nanosecond lasers and dielectrics interaction [60], [61], while also defect-assisted multiphoton absorption was described [62], [63].

Laser ablation of glass with nanosecond lasers has been demonstrated by several authors and different processes using the direct laser ablation or several workarounds such as laser-induced backside wet etching (LIBWE), laser-induced backside dry etching (LIBDE), laser-induced plasma-assisted ablation (LIPAA), or laser etching at a surface adsorbent layer (LESAL).

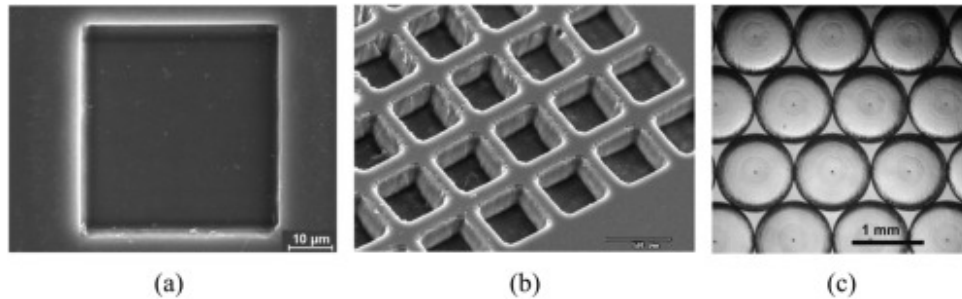
3.2. Direct laser ablation

As mentioned above, ablating transparent materials with nanosecond lasers is more challenging as compared to ultrashort pulsed lasers. The single pulse ablation threshold of fused silica was determined to $3.60\text{J}/\text{cm}^2$ for a 1030nm femtosecond (500fs) and $1054\text{J}/\text{cm}^2$ for a 1064nm nanosecond (20ns) laser [10]. However, it has been shown that through holes can be fabricated by helical drilling from the materials backside with a nanosecond laser in 4mm thick soda-lime glass using fluences in the order of tens of J/cm^2 [64]. Both the laser fundamental wavelength of 1064nm and its second harmonic of 532nm have been used, while for the infrared wavelength, fluences less than $38\text{J}/\text{cm}^2$ resulted in an unstable drilling process. Furthermore, all holes revealed cracks, being typical for long-pulse laser ablation being avoided by using ultrashort pulsed laser systems [14], [15]. The ablation of microchannels in soda-lime glass by a 20ns IR laser has been demonstrated using a fluence of $190\text{J}/\text{cm}^2$, while their roughness can be further reduced by thermal treatment for microfluidic applications [65]. Using short-wavelength lasers having wavelengths of 308, 248, and 193nm, ablation of fused silica has been studied with the lowest threshold fluence observed for the shortest wavelength [59], [60]. For 308nm, a fluence of about $20\text{J}/\text{cm}^2$ for polished fused silica is necessary, which is higher than that for 248nm, being about $10\text{J}/\text{cm}^2$ and about $3\text{J}/\text{cm}^2$ for the frontside for a wavelength of 193nm. A dual excimer-laser setup consisting of a KrF and a F_2 laser has shown to ablate fused silica with significantly lower fluences [66]. The F_2 laser with a small fluence of $60\text{mJ}/\text{cm}^2$ is used to create an excited state for the KrF laser for which a small fluence of $1.4\text{J}/\text{cm}^2$ was determined as the ablation threshold. Furthermore, the quality was significantly improved for the double-laser setup compared to the single KrF laser ablation, resulting in a higher quality without debris.

3.3. Laser-induced backside wet etching

LIBWE is used to ablate transparent materials with nanosecond lasers at low fluences as compared to direct nanosecond laser ablation [67], [68]. Typically, UV nanosecond lasers are focused through the transparent material on its backside, which is placed in a liquid absorber to ablate it from the backside, while the absorbing liquids are often solutions of pyrene in different liquids such as acetone, tetrahydrofuran, tetrachloroethylene, or toluene [68], [69], [70]. The mechanism underlying this process is based on the absorption of the laser energy in the liquid, which is intensively heated and transfers its thermal energy to the substrate material while both the hot substrate and occurring high pressure cause material removal [68], [71]. An etching threshold of $240\text{mJ}/\text{cm}^2$ for fused silica and acetone solution with pyrene as a liquid absorber has been found using a 30ns KrF laser, which is significantly lower as compared to direct laser ablation using nanosecond lasers [68]. It has also been shown that the achieved etching rate for the LIBWE process depends on the pyrene concentration as well as the applied laser fluence [67], [68]. The use of

a higher number of pulses linearly increases the etched depth [68], while for fused silica, typical etching rates of up to several 10nm/pulse have been achieved for different pyrene-containing solutions [68], [69]. Furthermore, two different fluence regimes have been reported with etching rates up to 25nm/pulse for the low and up to 200nm/pulse for the high fluences for fused silica [67]. In addition, liquid metal gallium has been used as an absorber to increase the etching rate up to about 1 μm /pulse, while also the ablation threshold was increased to about 1.3J/cm² [72]. Fig. 4A shows a 50×50 μm^2 etched crater in fused silica using liquid gallium and 30 pulses. Employing a thermoplasmonic LIBWE (TM-LIBWE) process with silver nanoparticles generated from an AgNO₃ precursor has shown an increasing etching rate of several μm /pulse on sapphire with a 527nm laser wavelength with 5ns pulses [73]. Furthermore, besides applying nanosecond lasers for the LIBWE process, also femto- and picosecond lasers have been demonstrated [74], [75]. Among using various experimental setups, this technique has been demonstrated to fabricate different types of structures such as blind holes in fused silica as shown in Fig. 4B [67] or the fabrication of micro lens arrays in quartz with their losses being about 13% including both reflection and absorption/scattering [69].



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Fig. 4. (A) 50 × 50 μm^2 etched square in fused silica using gallium as absorbing layer [72], (B) array of blind holes in fused silica [67], (C) microlens array fabricated with LIBWE in fused silica [69].

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3.4. Laser-induced backside dry etching

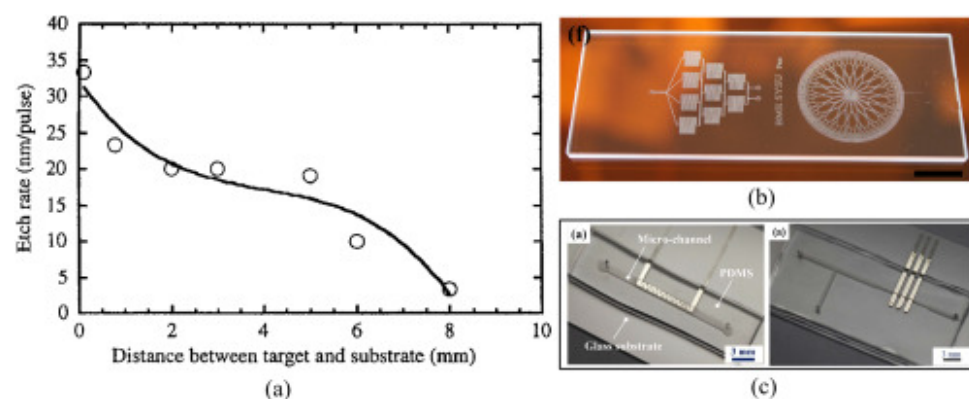
A similar process to LIBWE is the LIBDE using a solid absorber on the backside of the transparent materials instead of a liquid. Here, typically, a thin metal layer is coated on the backside of the transparent substrate and irradiated by a UV nanosecond laser [76], [77]. The laser energy is absorbed by the thin metal layer that is heated up and transfers its heat energy to the transparent substrate, causing melting or boiling with material ejection being caused by recoil forces of the vaporized metal in conjunction with the heated substrate [76], [78]. The ablation threshold for fused silica using a 30ns KrF laser with its wavelength of 248nm was found to be about 200mJ/cm² using a 100-nm-thick silver layer [77]. Comparing aluminum, silver, and copper as applied coating materials has shown the lowest value for aluminum, which is somehow higher for silver, whereas copper reveals a significantly increased threshold [78]. Also, variations of the typical LIBDE process using polymers as the absorption layer in air as well as the liquid environment [79] or a metal layer between two photoresist layers and the usage of infrared nanosecond lasers [80] have been demonstrated.

While the LIBDE process was primarily referred to a single-pulse ablation technique [76], [77], multipulse experiments have shown to increase the ablation depth with subsequent pulses following a saturation behavior [78], [81]. As the first pulse removes the metal layer, material removal with the subsequent pulses is attributed to residual metal in the upper resolidified layer, increasing the absorptivity of the substrate [78]. The use of different lasers with different wavelengths has shown that the pulse number for saturation strongly depends on the laser wavelength [81]. While saturation after two and three pulses has been found for the 308nm XeCl and the 248nm KrF laser, the 193nm ArF laser shows this behavior estimated only after 30 pulses. This circumstance is described by the high photon energy of the ArF laser being enough for breaking the Si—O bonds of the ablated fused silica and further reaction of the free oxygen atoms with the metal forming new molecules and matrix. However, using shorter wavelengths shows an increased etching depth being 3060nm after the tenth pulse for the shortest and 170nm for the longest wavelength.

3.5. Laser-induced plasma-assisted ablation

Laser-induced plasma-assisted ablation (LIPAA) is a method to process transparent materials with nanosecond lasers. It consists of a setup where the laser is focused through the transparent sample on an absorbing target that, contrary to the LIBDE process,

is not coated on the material backside, but a small gap between the specimen and the absorbing target is applied [82], [83], [84]. The incident laser is absorbed by the target causing material removal and plasma generation, which is hitting the backside of the substrate interacting with the incident laser, resulting in ablation on the backside of the substrate [84]. As expected, the distance between the target and the substrate plays a crucial role. Fig. 5 shows the dependence of the etch rate of fused quartz on the gap distance for 266-nm Q-switched Nd:YAG laser with a pulse duration of 6 ns and a stainless-steel target [84]. The highest etch rate of 33.3 nm/pulse is found for the smallest gap distance of 0.02 mm, with the etch rate decreasing with increasing gap distance. For large distances, an altered ablation mechanism due to particle deposition on the backside and laser ablation thereon without direct interaction of the plasma is described, while for intermediate distances, a combination of both might be responsible. The ablation threshold for the fused quartz at the smallest gap is reported to be 0.50 J/cm^2 , which also increases with increasing gap distance. Furthermore, a decreasing etch rate at the same fluence as well as an increasing ablation threshold was found for longer laser wavelengths in the LIPAA process using Ag as the target material [86]. However, these results confirm that the LIPAA process is feasible for different nanosecond laser wavelengths ranging from the UV region to the IR region.



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Fig. 5. (A) Dependence of the etch rate on the gap distance between target and substrate material [84], (B) microfluidic device patterns [83], (C) micro-heater chip (left) and droplet-based electrofluidic device for electrochemical detection (right) [85].
 +((A) Reprinted with permission J. Zhang, K. Sugioka, K. Midorikawa, Laser-induced plasma-assisted ablation of fused quartz using the fourth harmonic of a Nd <https://doi.org/10.1007/s003390050819> ↗, Springer Nature and Copyright Clearance Center. (B) Reprinted with permission C. Pan, K. Chen, B. Liu, L. Ren, J. Wang, Q. Hu, L. Liang, J. Zhou, L. Jiang, Fabrication of micro-texture channel on glass by laser-induced plasma-assisted ablation and chemical corrosion for microfluidic devices, J. Mater. Process.

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Variations of the LIPAA process have shown the fabrication of microfluidic (Fig. 5B) [83] or electrofluidic devices such as microheater chips (Fig. 5C left) or droplet-based devices for electrochemical detection (Fig. 5C right) on soda-lime glass [85]. Besides the LIPAA process based on nanosecond pulse durations, short pulsed laser with 500ps pulse duration has been demonstrated to ablate silicate glass using a copper target [87]. Furthermore, modified processes such as dual-beam LIPAA using an ultrashort pulsed laser have been shown for the fabrication of optical elements in sapphire [88].

3.6. Laser etching at a surface adsorbed layer

Another backside process for the ablation of transparent materials is LESAL. Here, the laser is focused through the transparent sample being absorbed by a thin layer of hydrocarbons adsorbed on its backside, while the adsorbed layer is formed by the evaporation of liquid toluene in a chamber by an electrical heater [89], [90]. The ablation threshold found for fused silica using a 248nm and 30ns pulsed laser is 0.7J/cm² for 300 pulses [90]. Furthermore, the etch rate increases with increasing fluence and saturates at about 1.3nm/pulse between fluences of 2J/cm² and 5J/cm². For higher fluences, a strong increase in the etch rate is observed, which is linked to ablation processes being only influenced by the adsorbed material. However, the intermediate fluence range where the ablation rate saturates is very interesting due to the well-controlled ablation depth and a very low surface roughness [90], [91]. Thus, LESAL has shown to remove material leaving very smooth ablation crates with roughness clearly below 1 nm being in the range or somewhat higher than the initial substrate roughness [90], [91]. Due to the process-related small surface roughness, relief gratings with a periodicity of 3μm and a depth of 800nm can be fabricated into fused silica revealing a high quality [92].

4. Glass processing with CO₂ lasers

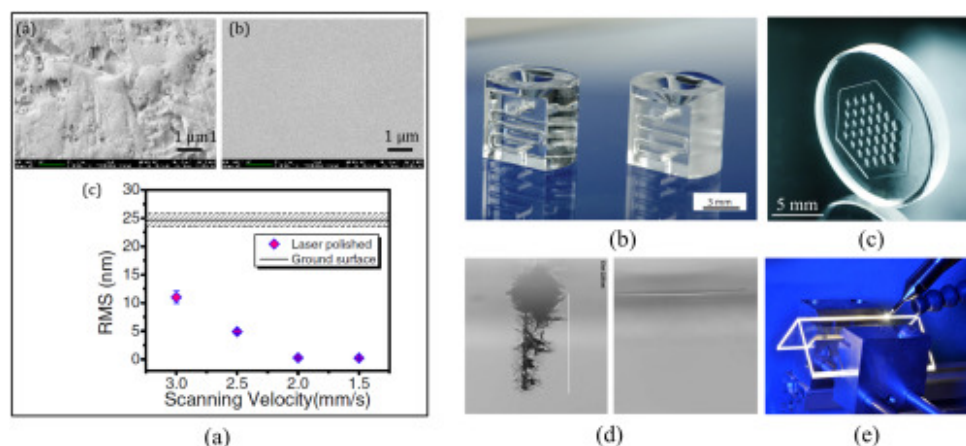
4.1. Material ablation mechanism

CO₂ lasers with their typical wavelength of 10.6μm have manifold been applied for glass processing, utilizing the high absorption of their M-IR wavelength of about, for example, 80% in fused silica with a corresponding small temperature-dependent penetration depth of 4–40μm for temperatures between 2000 and 300K [93], [94]. Due to their low photon energy of 0.2eV, the Si-O bonds with their bond energy of 4.6eV in glass cannot be broken and the laser energy is strongly absorbed by a Si—O bond vibration mode which is transferred to thermal energy by the excitation of lattice vibrations [95]. Therefore, using CO₂ lasers results in melting of the irradiated glass or causing material removal by further heating up to the material evaporation temperature.

4.2. Applications of CO₂ lasers

Based on the described absorption mechanism, CO₂ lasers are often used as heating sources for applications such as polishing [96], [97], laser-assisted reshaping [23], [98], laser welding [99], [100], or damage repair and mitigation [94], [101]. Laser polishing is achieved by heating the material, causing a viscosity reduction and flow of the melted glass by surface tension, while the temperature should not exceed the evaporation point of the irradiated material, almost avoiding material removal [93], [96]. Fig. 6A shows results of CO₂ laser polishing of a fine grounded fused silica sample [96]. The variation of the scanning speed reveals a significant influence on the polishing process, where the initial surface roughness of about 25nm can be further reduced by the CO₂ laser to 0.22nm. A microfluidic device fabricated by selective-laser etching in fused silica is shown in Fig. 6B before (right) and after (left) CO₂ laser polishing [93]. After the polishing process, the inner walls of the microfluidic channels are smoothened. Laser polishing has also been demonstrated for the fabrication of different optical elements using a CO₂ laser polishing process. Different lenses such as cylindrical- and dome-type lens arrays on fused silica [23] or hemispherical lens arrays in soda-lime glass [98] can be fabricated by CO₂ laser reshaping accompanied by a surface roughness reduction. Here, the microstructures are generated on the glass substrates by laser ablation, while the CO₂ laser reshaping is used to generate the optical geometry, which in turn limits the geometrical freedom. In contrast, Fig. 6C shows an axicon array where its geometry is introduced by precise ultrashort pulsed laser ablation and the CO₂ laser is used for polishing while a reshaping of the geometry should be minimized [97]. However, due to surface melting during the polishing process, smaller geometrical transformations are observed. Furthermore, the direct fabrication of lenses and lens arrays on Corning 7059 glass substrates has been demonstrated by a single CO₂ laser process, heating the surface to its working point while a hyperboloid lens geometry is formed by surface tension [102]. For this process, a glass with thermostable properties with a small expansion coefficient has to be chosen. Melting the irradiated substrates can also be used for damage mitigation as shown in Fig. 6D [94]. Here, a CO₂ laser having a wavelength of 10.6μm is compared to a frequency-doubled CO₂ laser having a wavelength of 4.6μm. It was found that

the absorption length for the 4.6 μm laser is in the range of 271–1397 μm for temperatures between 2000 and 300K and thus is significantly longer as compared to that of a 10.6 μm CO₂ laser. The damage track shown in Fig. 6D was generated in fused silica by a Q-switched Nd:YAG laser and has a depth of 225 μm (left). After being irradiated with the 4.6 μm CO₂ laser (right), the substrate damaged can be completely removed, while the damage track is filled with glass driven by surface tension, leaving a shallow crater on the surface. Another heat-driven process is the joining of two glass plates as shown in Fig. 6E by using an additional glass fiber as a filler [99].



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Fig. 6. (A) Scanning electron microscope images of fine ground (left) and CO₂ laser polished surface (right) and influence of the laser scanning velocity [96], (B) CO₂ laser polished (left) and unpolished (right) microfluidic device [93], (C) optical element polished with CO₂ laser after ultrashort pulsed laser ablation of an axicon array geometry [97], (D) damage in fused silica before (left) and after (right) 4.6 μm CO₂ laser healing [94], (E) CO₂ laser joining process of two glass plates with additional fiber [99]. ((A) Reprinted with permission T. He, C. Wei, Z. Jiang, Y. Zhao, J. Shao, Super-smooth surface demonstration and the physical mechanism of CO₂ laser polishing of fused silica, Opt. Lett. 43 (23) (2018) 5777–5780. <https://doi.org/10.1364/OL.43.005777> ↗, The Optical Society. (B) Reprinted with permission C. Weingarten, S. Steenhusen, M. Hermans, E. Willenborg, J.H. Schleifenbaum, Laser polishing and 2PP structuring of inside microfluidic channels in fused silica, Microfluid. Nanofluid. 21 (11) (2017) 165. <https://doi.org/10.1007/s10404-017-2000-x> ↗, Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/> ↗). (C) Reprinted with permission S. Schwarz, S. Rung, C. Esen, R. Hellmann,

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